SG No. 176 C_{6h}^2 $P6_3/m$ [hexagonal]

* plus set: $+[0,\ 0,\ 0]$

* Wyckoff site: `2a`, site symmetry: `-6..`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `-6..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,-10,-11,-12]`
2	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[4,5,6,-7,-8,-9]`

Table 2: Wyckoff bond: `6b@2a`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,10]`
2	$[-Y,\ X-Y,\ 0]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[2,11]`
3	$[-X+Y,\ -X,\ 0]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[3,12]`
4	$[-X,\ -Y,\ 0]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[4,7]`
5	$[Y,\ -X+Y,\ 0]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[5,8]`
6	$[X-Y,\ X,\ 0]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[6,9]`

Table 3: Wyckoff bond: `12c@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[6]`
7	$[-X,\ -Y,\ -Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[7]`
8	$[Y,\ -X+Y,\ -Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[8]`
9	$[X-Y,\ X,\ -Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[9]`
10	$[X,\ Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[10]`
11	$[-Y,\ X-Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[11]`
12	$[-X+Y,\ -X,\ -Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[12]`

* Wyckoff site: `2b`, site symmetry: `-3..`

Table 4: Wyckoff bond: `2a@2b`, bond symmetry: `-3..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[0,\ 0,\ 0]$	`[1,2,3,-7,-8,-9]`
2	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,5,6,-10,-11,-12]`

Table 5: Wyckoff bond: `6b@2b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1,-7]`
2	$[-Y, X-Y, Z]$	$[0, 0, 0]$	`[2,-8]`
3	$[-X+Y, -X, Z]$	$[0, 0, 0]$	`[3,-9]`
4	$[-X, -Y, Z]$	$[0, 0, \frac{1}{2}]$	`[4,-10]`
5	$[Y, -X+Y, Z]$	$[0, 0, \frac{1}{2}]$	`[5,-11]`
6	$[X-Y, X, Z]$	$[0, 0, \frac{1}{2}]$	`[6,-12]`

* Wyckoff site: `2c`, site symmetry: `-6..`

Table 6: Wyckoff bond: `2a@2c`, bond symmetry: `-6..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}]$	`[1,2,3,-10,-11,-12]`
2	$[0, 0, Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}]$	`[4,5,6,-7,-8,-9]`

Table 7: Wyckoff bond: `6b@2c`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}]$	`[1,10]`
2	$[-Y, X-Y, 0]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}]$	`[2,11]`
3	$[-X+Y, -X, 0]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}]$	`[3,12]`
4	$[-X, -Y, 0]$	$[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}]$	`[4,7]`
5	$[Y, -X+Y, 0]$	$[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}]$	`[5,8]`
6	$[X-Y, X, 0]$	$[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}]$	`[6,9]`

Table 8: Wyckoff bond: `12c@2c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}]$	`[1]`
2	$[-Y, X-Y, Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}]$	`[2]`
3	$[-X+Y, -X, Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}]$	`[3]`
4	$[-X, -Y, Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}]$	`[4]`
5	$[Y, -X+Y, Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}]$	`[5]`
6	$[X-Y, X, Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}]$	`[6]`
7	$[-X, -Y, -Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}]$	`[7]`
8	$[Y, -X+Y, -Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}]$	`[8]`
9	$[X-Y, X, -Z]$	$[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}]$	`[9]`
10	$[X, Y, -Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}]$	`[10]`
11	$[-Y, X-Y, -Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}]$	`[11]`
12	$[-X+Y, -X, -Z]$	$[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}]$	`[12]`

* Wyckoff site: `2d`, site symmetry: `-6..`

Table 9: Wyckoff bond: `2a@2d`, bond symmetry: `-6..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[1,2,3,-10,-11,-12]`
2	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[4,5,6,-7,-8,-9]`

Table 10: Wyckoff bond: `6b@2d`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[1,10]`
2	$[-Y,\ X-Y,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[2,11]`
3	$[-X+Y,\ -X,\ 0]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[3,12]`
4	$[-X,\ -Y,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[4,7]`
5	$[Y,\ -X+Y,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[5,8]`
6	$[X-Y,\ X,\ 0]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[6,9]`

Table 11: Wyckoff bond: `12c@2d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[1]`
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[2]`
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[3]`
4	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[4]`
5	$[Y,\ -X+Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[5]`
6	$[X-Y,\ X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[6]`
7	$[-X,\ -Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[7]`
8	$[Y,\ -X+Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[8]`
9	$[X-Y,\ X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[9]`
10	$[X,\ Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[10]`
11	$[-Y,\ X-Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[11]`
12	$[-X+Y,\ -X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[12]`

* Wyckoff site: `4e`, site symmetry: `3..`

Table 12: Wyckoff bond: `4a@4e`, bond symmetry: `3..`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[0,\ 0,\ z]$	`[1,2,3]`
2	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[4,5,6]`
3	$[0,\ 0,\ -Z]$	$[0,\ 0,\ -z]$	`[7,8,9]`
4	$[0,\ 0,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[10,11,12]`

Table 13: Wyckoff bond: `12b@4e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[0,\ 0,\ z]$	`[1]`

continued ...

Table 13

No.	vector	center	mapping
2	$[-Y,\ X-Y,\ Z]$	$[0,\ 0,\ z]$	[2]
3	$[-X+Y,\ -X,\ Z]$	$[0,\ 0,\ z]$	[3]
4	$[-X,\ -Y,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[4]
5	$[Y,\ -X+Y,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[5]
6	$[X-Y,\ X,\ Z]$	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[6]
7	$[-X,\ -Y,\ -Z]$	$[0,\ 0,\ -z]$	[7]
8	$[Y,\ -X+Y,\ -Z]$	$[0,\ 0,\ -z]$	[8]
9	$[X-Y,\ X,\ -Z]$	$[0,\ 0,\ -z]$	[9]
10	$[X,\ Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[10]
11	$[-Y,\ X-Y,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[11]
12	$[-X+Y,\ -X,\ -Z]$	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[12]

* Wyckoff site: 4f, site symmetry: 3..

Table 14: Wyckoff bond: 4a@4f, bond symmetry: 3..

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	[1,2,3]
2	$[0,\ 0,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	[4,5,6]
3	$[0,\ 0,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	[7,8,9]
4	$[0,\ 0,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	[10,11,12]

Table 15: Wyckoff bond: 12b@4f, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	[1]
2	$[-Y,\ X-Y,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	[2]
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	[3]
4	$[-X,\ -Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	[4]
5	$[Y,\ -X+Y,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	[5]
6	$[X-Y,\ X,\ Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ z+\frac{1}{2}\right]$	[6]
7	$[-X,\ -Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	[7]
8	$[Y,\ -X+Y,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	[8]
9	$[X-Y,\ X,\ -Z]$	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	[9]
10	$[X,\ Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	[10]
11	$[-Y,\ X-Y,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	[11]
12	$[-X+Y,\ -X,\ -Z]$	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}-z\right]$	[12]

* Wyckoff site: 6g, site symmetry: -1

Table 16: Wyckoff bond: 6a@6g, bond symmetry: -1

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{2},\ 0,\ 0\right]$	[1,-7]
2	$[-Y,\ X-Y,\ Z]$	$\left[0,\ \frac{1}{2},\ 0\right]$	[2,-8]
3	$[-X+Y,\ -X,\ Z]$	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[3,-9]

continued ...

Table 16

No.	vector	center	mapping
4	$[-X, -Y, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[4,-10]`
5	$[Y, -X+Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[5,-11]`
6	$[X-Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[6,-12]`

* Wyckoff site: `6h`, site symmetry: `m..`

Table 17: Wyckoff bond: `6a@6h`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[x, y, \frac{1}{4}\right]$	`[1,10]`
2	$[-Y, X-Y, 0]$	$\left[-y, x-y, \frac{1}{4}\right]$	`[2,11]`
3	$[-X+Y, -X, 0]$	$\left[-x+y, -x, \frac{1}{4}\right]$	`[3,12]`
4	$[-X, -Y, 0]$	$\left[-x, -y, \frac{3}{4}\right]$	`[4,7]`
5	$[Y, -X+Y, 0]$	$\left[y, -x+y, \frac{3}{4}\right]$	`[5,8]`
6	$[X-Y, X, 0]$	$\left[x-y, x, \frac{3}{4}\right]$	`[6,9]`

Table 18: Wyckoff bond: `6b@6h`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[x, y, \frac{1}{4}\right]$	`[1,-10]`
2	$[0, 0, Z]$	$\left[-y, x-y, \frac{1}{4}\right]$	`[2,-11]`
3	$[0, 0, Z]$	$\left[-x+y, -x, \frac{1}{4}\right]$	`[3,-12]`
4	$[0, 0, Z]$	$\left[-x, -y, \frac{3}{4}\right]$	`[4,-7]`
5	$[0, 0, Z]$	$\left[y, -x+y, \frac{3}{4}\right]$	`[5,-8]`
6	$[0, 0, Z]$	$\left[x-y, x, \frac{3}{4}\right]$	`[6,-9]`

Table 19: Wyckoff bond: `12c@6h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, y, \frac{1}{4}\right]$	`[1]`
2	$[-Y, X-Y, Z]$	$\left[-y, x-y, \frac{1}{4}\right]$	`[2]`
3	$[-X+Y, -X, Z]$	$\left[-x+y, -x, \frac{1}{4}\right]$	`[3]`
4	$[-X, -Y, Z]$	$\left[-x, -y, \frac{3}{4}\right]$	`[4]`
5	$[Y, -X+Y, Z]$	$\left[y, -x+y, \frac{3}{4}\right]$	`[5]`
6	$[X-Y, X, Z]$	$\left[x-y, x, \frac{3}{4}\right]$	`[6]`
7	$[-X, -Y, -Z]$	$\left[-x, -y, \frac{3}{4}\right]$	`[7]`
8	$[Y, -X+Y, -Z]$	$\left[y, -x+y, \frac{3}{4}\right]$	`[8]`
9	$[X-Y, X, -Z]$	$\left[x-y, x, \frac{3}{4}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[x, y, \frac{1}{4}\right]$	`[10]`
11	$[-Y, X-Y, -Z]$	$\left[-y, x-y, \frac{1}{4}\right]$	`[11]`
12	$[-X+Y, -X, -Z]$	$\left[-x+y, -x, \frac{1}{4}\right]$	`[12]`

* Wyckoff site: `12i`, site symmetry: `1`

Table 20: Wyckoff bond: `12a@12i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$[x,\, y,\, z]$	`[1]`
2	$[-Y,\, X-Y,\, Z]$	$[-y,\, x-y,\, z]$	`[2]`
3	$[-X+Y,\, -X,\, Z]$	$[-x+y,\, -x,\, z]$	`[3]`
4	$[-X,\, -Y,\, Z]$	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	`[4]`
5	$[Y,\, -X+Y,\, Z]$	$\left[y,\, -x+y,\, z+\frac{1}{2}\right]$	`[5]`
6	$[X-Y,\, X,\, Z]$	$\left[x-y,\, x,\, z+\frac{1}{2}\right]$	`[6]`
7	$[-X,\, -Y,\, -Z]$	$[-x,\, -y,\, -z]$	`[7]`
8	$[Y,\, -X+Y,\, -Z]$	$[y,\, -x+y,\, -z]$	`[8]`
9	$[X-Y,\, X,\, -Z]$	$[x-y,\, x,\, -z]$	`[9]`
10	$[X,\, Y,\, -Z]$	$\left[x,\, y,\, \frac{1}{2}-z\right]$	`[10]`
11	$[-Y,\, X-Y,\, -Z]$	$\left[-y,\, x-y,\, \frac{1}{2}-z\right]$	`[11]`
12	$[-X+Y,\, -X,\, -Z]$	$\left[-x+y,\, -x,\, \frac{1}{2}-z\right]$	`[12]`